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Buchwald–Hartwig amination of aryl esters and chlorides catalyzed by the dianisole-decorated Pd–NHC complex†

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A modular and generic method for the Buchwald–Hartwig amination reactions of relatively unreactive aryl esters as acyl electrophiles and aryl chlorides as aryl electrophiles has been developed, leading to the efficient synthesis of amides/amines under air conditions and with low catalyst loadings. The success of this catalytic protocol is mainly attributed to the modification of the Pd–IPr skeleton with sterically hindered and electron-donating anisole groups. This method also features good functional group tolerance and excellent chemoselectivities. In summary, the results presented herein suggest the possibility of developing a versatile and general protocol for diverse electrophiles to undergo the Buchwald-Hartwig amination reactions, avoiding too much consideration of the reaction conditions for the substrate-dependent C–N bond formations.

Introduction

The palladium-catalyzed Buchwald–Hartwig amination (BHA) reaction has been demonstrated to be one of the most versatile synthetic tools for carbon–nitrogen bond formation.[1] Traditionally, aryl iodides/bromides are employed as electrophiles in this reaction, and many elegant and efficient catalytic systems have been reported based on these substrates. To date, the reaction has been extended to a variety of electrophiles, such as aryl triflates,[2] tosylates,[3] sulfonates,[4] esters,[5] chlorides,[6] and mesylates,[7] and each electrophile needs its specific amine coupling partner, ligand system and other reaction conditions. It is noteworthy that among all these electrophiles, aryl esters and chlorides are attractive because they are readily available, bench-stable, and inexpensive.[5,6] However, the employment of aryl esters as cross-coupling electrophiles could cause problems with selective C–O activation, as either the acyl C–O bond or the aryl C–O bond could be potentially cleaved.[5,8] Furthermore, due to the relatively low reactivities of aryl esters and chlorides, the oxidative additions of C–O and C–Cl bonds to the reactive palladium center are rather unfavorable compared to C–Br or C–I bonds.[9] To the best of our knowledge, so far, there is no precedent of using a universal catalytic system for BHA reactions of both aryl esters as acyl electrophiles and aryl chlorides as aryl electrophiles under mild aerobic conditions and with excellent chemoselectivities. Therefore, developing an efficient and general catalytic protocol that enables the rapid and modular synthesis of C–N bonds through aryl ester and chloride Buchwald–Hartwig reactions is highly desired and sought after.

The first example of palladium-catalyzed aryl ester BHA reactions was reported by Newman and co-workers in 2017,[5a] employing the highly reactive palladium N-heterocyclic carbene (Pd–NHC) complex Pd(IPr)(allyl)Cl as the precatalyst. At about the same time, Szostak *et al.* reported a versatile Pd–PEPPSI–IPr (PEPPSI, pyridine-enhanced precatalyst preparation, stabilization, and initiation) precatalyst,[5b] which could efficiently promote both the amidation of aryl esters and the transamidation of amides under the same reaction conditions. Moreover, the same group reported another example using Pd(IPr)(acac)Cl as an efficient precatalyst.[5c] Apart from these elegant examples, the groups of Mohadjer Beromi[5d] and Mendoza-Espinosa[5e] reported highly effective BHA reactions of aryl esters employing Pd(SIPr)(η^3-1-^{t}Bu-indenyl)Cl and Pd(NHC)(cinnamyl)Cl, respectively, as precatalysts. Of note, most of these reported transformations used saturated/unsaturated IPr as the ancillary ligands, and different types of throwaway ligands for Pd–NHC have been employed without any definite preference.[5] Therefore, we were interested in investigating the catalytic performances of the modified Pd–PEPPSI–IPr complexes in the aryl ester BHA reactions. Considering that it has been established that sterically hindered ancillary ligands

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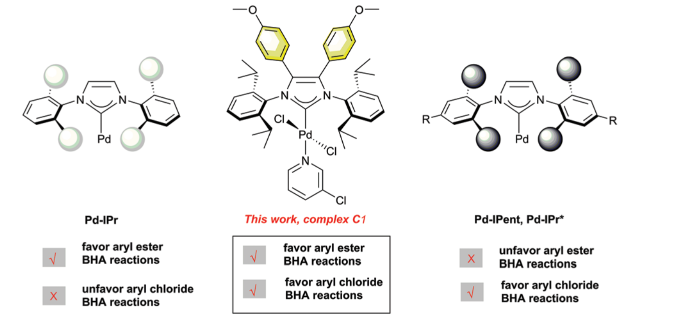
could facilitate reductive elimination[10] and the steric bulk at the 2,6-positions of *N*-aryl moieties helps stabilize the active Pd(0) species[11] and achieve efficient couplings under aerobic conditions,[11f–l] it is reasonable to assume that the more bulky Pd–PEPPSI–IPent should lead to improved conversion. Unfortunately, as indicated in Szostak's report,[5b] the use of Pd–PEPPSI–IPent resulted in much less efficient coupling, which may be due to the greater steric repulsion of the bulkier substituents on *N*-aryl moieties that affects the effective oxidative addition of C–O to the palladium center. We wondered whether the modification of the backbone of IPr with bulky substitutions, instead of increasing the steric bulk of the *N*-aryl moieties, could help circumvent this problem and make the amination of aryl esters more synthetically accessible in terms of air conditions and lower catalyst loadings.

Different from the aryl ester BHA reactions, tremendous progress has been made with regard to the aryl chloride Buchwald–Hartwig reactions.[6] In these reactions, the modification of the scaffold of the NHC ligands in the Pd–PEPPSI precatalysts resulted in a practical and cost-effective way to construct mono-aminated products.[11h,j,l,12] Also, it has been revealed that the steric hindrance resulting from the substitutions of the precatalysts, such as Pd–PEPPSI–IPent^{Cl} and Pd–PEPPSI–IPr$^{(NMe_2)_2}$,[12d–g] could not only prevent a second amination but also facilitate the reductive elimination, thereby suppressing the undesirable β-H elimination. Also, given that the enhanced electron donation from the NHC ligands to the palladium center may be beneficial for the oxidative addition,[10] it is reasonable to expect that the modifications of the NHC skeletons by introducing bulky groups with strong electron-donating capacities may result in even higher reactivities and selectivities toward mono-amination products. Furthermore, considering that the framework of Pd–IPr may be vital for the effective amination of aryl esters and as part of our continued interest in the aryl ester and chloride Buchwald–Hartwig reactions, we envisioned that the modification of the skeleton of IPr with sterically demanding and electron-rich groups might provide an effective and general catalytic protocol for aryl ester and chloride Buchwald–Hartwig reactions.

We recently demonstrated that 1,3,4,5-tetraarylimidazolylidene based NHC ligands, in which two aryl groups are attached to the NHC backbone, exhibit much stronger σ-donating ability and are bulkier than the normal 1,3-diarylimidazolylidene ligands.[11f,12i] They show extraordinarily high catalytic activities in the direct arylation of imidazoles with aryl bromides[11f] and in the Suzuki–Miyaura reaction of aryl chlorides under air conditions.[12i] Therefore, we were motivated to investigate whether the 1,3,4,5-tetraarylimidazolylidene skeleton of NHCs has an even more pronounced effect on the BHA reactions. Herein, we explored the catalytic ability of Pd–PEPPSI–IPr$^{(PhOMe)_2}$ (**C1**,[12i] as shown in Scheme 1), in which the IPr ligand was modified by introducing two –PhOMe groups at the imidazolyl ring, toward the Buchwald–Hartwig aminations. The results demonstrated that the bulky steric hindrance and high electron-donating properties of the backbone modified NHC ligand indeed resulted in a significant increase in the catalytic properties. As such, the amination reactions of aryl esters and chlorides can be conducted under one set of conditions with a broad substrate scope, excellent yields, and high chemoselectivities, thus suggesting the possibility of developing a general protocol for diverse electrophiles to undergo the Buchwald–Hartwig aminations.

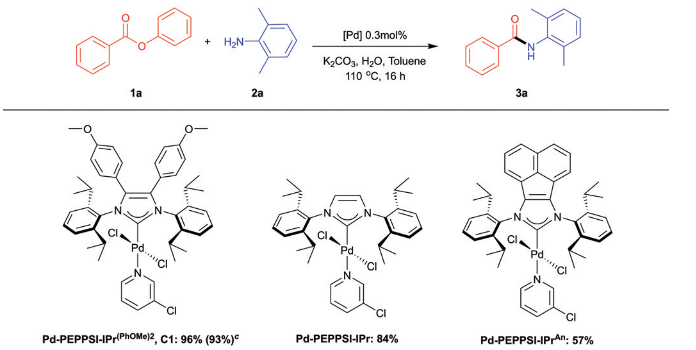


Scheme 1 The design strategy of Pd–NHCs for aryl ester and chloride amination reactions.

Results and discussion

The Pd–PEPPSI–NHC complexes investigated in this study are shown in Table 1. To evaluate the catalytic performances of these complexes in the BHA reactions, aryl ester phenyl benzoate (**1a**) and sterically-hindered poorly nucleophilic 2,6-dimethyl-aniline (**2a**) were initially chosen as model substrates to start our investigation. As shown in Table 1, the reactions were conducted under the following conditions: 0.3 mol% of palladium loading, 1.5 equivalents of K_2CO_3, H_2O (10 mol%), and 3 mL of toluene with heating at 110 °C for 16 h under air, based on previously established reaction conditions by Newman.[5a] To our delight, the results showed that **C1** was remarkably active and the most effective precatalyst among the

Table 1 Screening of different framework catalysts for the aryl ester amination reaction[a,b]



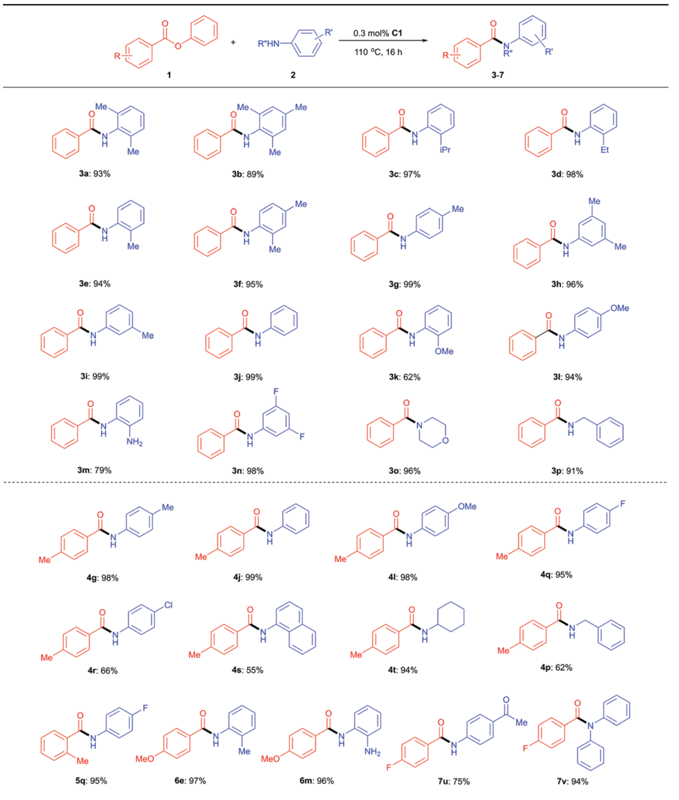
[a] Reagents and conditions: phenyl benzoate (1 mmol), 2,6-dimethylaniline (1.2 mmol), palladium complex (0.3 mol%), K_2CO_3 (1.5 mmol), H_2O (10 mol%), and toluene (3 mL) in air. [b] Cross-coupling product determined by GC, average of two runs.

three, affording the desired acyl C–O activation product **3a** in an almost quantitative yield of 96% with an extremely low catalyst loading.[5] The commonly employed precatalyst for the cross-coupling reaction, Pd–PEPPSI–IPr, provided the desired product in 84% yield. Intriguingly, Pd–PEPPSI–IPr^{An}, with a σ-donating acenaphthyl group, was proved to be the least efficient, affording the desired product in an unsatisfactory yield of 57%. All these results demonstrated that the strategy of modifying the NHC backbone within the Pd–PEPPSI–IPr framework worked well. Notably, the calculated percent buried volume ($\%V_{Bur}$) value of **C1** is 38.7, which is much higher than the values of Pd–IPr (34.3) and Pd–IPr^{An} (34.7),[12i] indicating its bulky steric hindrance. The Tolman electronic parameter (TEP) value of **C1** estimated from $Ir(L)(CO)_2Cl$ is 2043.1 cm^{-1}, showing its higher σ-donating ability than that of IPr (2050.2 cm^{-1}) and which is slightly lower than that of IPr^{An} (2041.8 cm^{-1}).[12i] Thus, the superior activity of **C1** may be due to the synergistic effect of the bulky steric hindrance and the strong electron-donating properties of the –PhOMe groups, which may facilitate the oxidative addition of the C(acyl)–O bond with the Pd–NHC complex and the reductive elimination of the coupling product. After a systematic survey of the bases, solvents, and additives (see Table S1 in the ESI†), the optimized reaction conditions were established as follows: phenyl benzoate (1.0 equiv.), amine (1.2 equiv.), **C1** (0.3 mol%) as the precatalyst, K_2CO_3 (1.5 equiv.) as the base, and H_2O (10 mol%) in 3 mL of toluene at 110 °C for 16 h in an aerobic environment. It is noteworthy that without the presence of the catalyst, no reaction was observed (see Table S1 in the ESI†).

With the optimized conditions established, we then turned our attention to examine the generality of the Pd–PEPPSI–IPr$^{(PhOMe)_2}$-catalyzed amination of aryl esters. As shown in the upper part of Table 2, a wide range of aromatic and aliphatic amines smoothly reacted with phenyl benzoate, providing rapid access to various amides. Anilines with electron-withdrawing, electron-donating, and electron-neutral substitutions are all compatible with the developed coupling conditions, affording the corresponding amide products (**3a** to **3j**, **3l**, and **3n**) in good to excellent yields. Sterically-hindered substrates, even 2,6-substituted anilines, can also be smoothly coupled, providing the desired products (**3a** and **3b**) in excellent yields of 89–93%. However, the substrate having an electron-donating –OMe group at the *ortho* position of the aniline resulted in a moderate product output (**3k**, 62% yield). Notably, benzene-1,2-diamine can also selectively couple with phenyl benzoate, furnishing the highly chemoselective mono-amination product **3m** in 79% yield. The cyclic secondary aliphatic amine morpholine could also be readily coupled to deliver the corresponding product **3o** smoothly in a yield of 96%. The primary amine phenylmethanamine, which could easily undergo the β-H elimination, can efficiently couple with phenyl benzoate, furnishing the monoaminated product **3p** with high selectivity (91% yield).

To further probe the substrate scope of the reaction, we examined a series of substituted aryl esters with a broader range of amines. As shown in the lower part of Table 2, the reaction also exhibits a broad substrate scope and high functional group tolerance toward the aryl ester components. Numerous substituted aryl esters, regardless of the steric effect and electronic nature of the substituents, are all amenable to the reaction conditions, providing a practical access to various amides (**4g** to **7v**). With regard to the amine components, no clear correlation was observed between the electronic properties of the substituents and the yields of the products (as shown in **4l** and **4q**). Anilines with sterically hindered substituents (as shown in **6e**, **6m** and **7v**) were all well tolerated. However, a relatively lower yield (**4s**, 55%) was obtained when sterically hindered 1-naphthalen-amine was used as a nucleophile. Additionally, a low yield of the product (less than 10%) was obtained by GC analysis when carbazole was used as the coupling partner with phenyl benzoate, and the attempts to isolate the corresponding product failed. Aliphatic primary amines can also be coupled successfully, furnishing the coupled products (**4t** and **4p**) in moderate to excellent yields (62–94%). It is noteworthy that the aniline substrate bearing a base-sensitive substituent is also amenable to the reaction conditions, affording the desired product **7u** in 75% yield. Considering the mild reaction conditions and the complexity of the products presented in Table 2, our methodology demon-

Table 2 Aryl ester Buchwald–Hartwig amination reactions catalysed by **C1**[a,b]



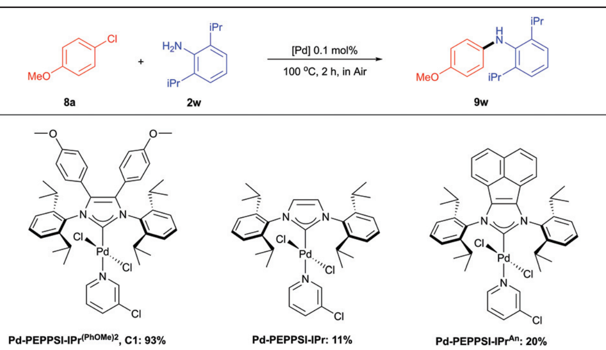
[a] Reagents and conditions: phenyl benzoate (1 mmol), amine (1.2 mmol), Pd–PEPPSI–IPr$^{(PhOMe)_2}$ (0.3 mol%), K_2CO_3 (1.5 mmol), H_2O (10 mol%), and toluene (3 mL) in air. [b] Isolated yields, average of two runs.

strated the significant advantages of "air-stable catalyst performed in air" and comprehensive substrate scope,[5] regardless of the electronics of the amine nucleophiles.

With an efficient protocol for the aryl ester amination in hand, we envisioned that the extraordinary catalytic abilities of complex **C1** might also enable highly efficient C–N amination of aryl chlorides under similar conditions. This methodology, if it works, would attest to the generality of the developed catalytic system for the formation of C–N bonds. Thus, the most challenging combination of coupling partners, the electron-rich aryl chloride 4-chloroanisole (**8a**) and sterically hindered poorly nucleophilic 2,6-diisopropylaniline (**2w**), was first selected to test the catalytic potential of the Pd–PEPPSI complexes. As we expected, the use of complex **C1** resulted in an outstanding performance again, delivering the desired coupling product **9w** in an almost quantitative yield (93%) with only 0.1 mol% catalyst loading (as shown in Table 3). The use of both Pd–PEPPSI–IPr and Pd–PEPPSI–IPr^{An} resulted in much lower yields, 11% and 20%, respectively, which demonstrated that the steric nature of the palladium precatalysts played a significant role in the reaction. With these initial promising results, a systematic study of different bases and solvents was conducted (see Table S2 in the ESI†). The optimized reaction conditions were then established as follows: aryl chloride (1.0 equiv.), amine (1.2 equiv.), **C1** (0.1 mol%) as the precatalyst, and KO^{t}Am (1.5 equiv.) as the base in 4 mL of dioxane at 100 °C for 2 h in an aerobic environment.

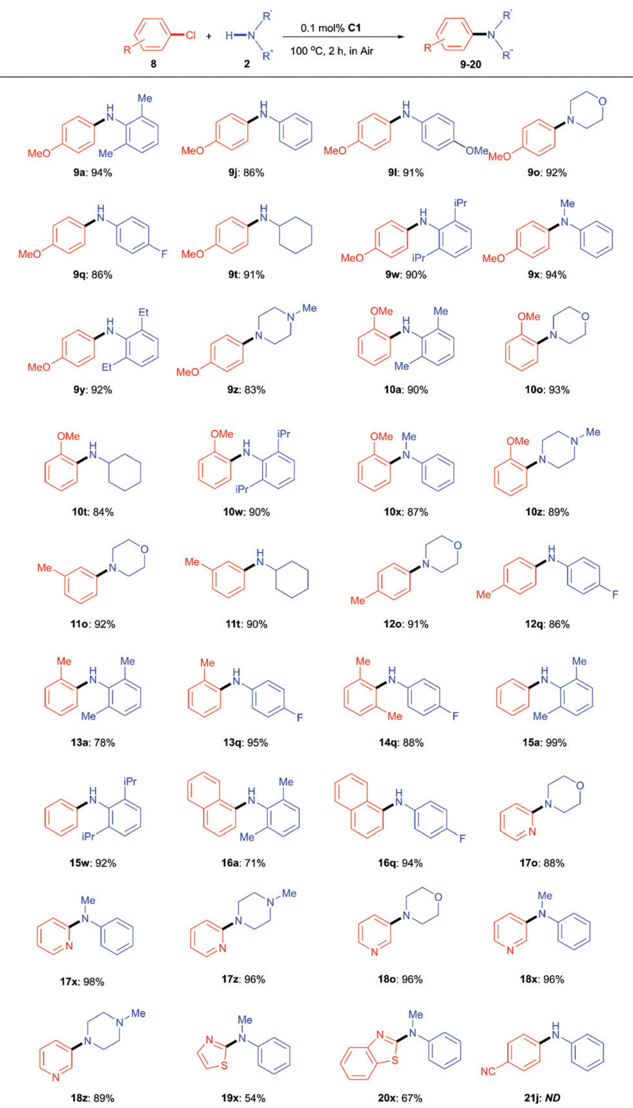
With a defined catalytic system, a wide range of (hetero)aryl chlorides with primary and secondary amines were investigated (Table 4) to further explore the uses of the developed methodology. With regard to the aryl chloride components, substitutions of the electron-donating and electron-neutral groups at different positions on the aryl ring did not affect the efficiencies of the catalytic process, affording the corresponding products (**9a** to **20x**) in good to excellent yields (54–99% yields). However, an extremely low yield was observed when an electron-withdrawing group was present (**21j**) and the

Table 3 Screening of different framework catalysts for the aryl chloride amination reaction[a,b]



[a] Reagents and conditions: 4-chloroanisole (1 mmol), 2,6-diisopropylaniline (1.2 mmol), palladium complex (0.1 mol%), KO^{t}Am (1.5 mmol), and 1,4-dioxane (4 mL) in air. [b] Cross-coupling product determined by GC, average of two runs.

Table 4 Aryl chloride Buchwald–Hartwig amination reactions catalysed by **C1**[a,b]



[a] Reagents and conditions: aryl chloride (1 mmol), amine (1.2 mmol), Pd–PEPPSI–IPr$^{(PhOMe)_2}$ (0.1 mol%), KO^{t}Am (1.5 mmol), 1,4-dioxane (4 mL) in air. [b] Isolated yields, average of two runs.

attempts to isolate the product failed. A number of heteroaryl chlorides also worked well, providing the corresponding *N*-arylation products (**17o–20x**) in 54%–98% yields.

To our delight, in terms of the secondary amine coupling partners, aliphatic and aromatic secondary amines, such as morpholine, 1-methyl-piperazine, and *N*-methylaniline, were all amenable to the developed catalytic conditions, affording the corresponding products (**9o, 9x, 9z, 10o, 10x, 10z, 11o, 12o,** and **17o–20x**) in moderate to excellent yields (54–98%). Of particular note is that sterically hindered *N*-methylaniline coupled well with (hetero)aryl chlorides, leading to the *N*-(hetero)arylated products (**9x, 10x, 17x, 18x, 19x** and **20x**) in moderate to excellent yields (54–98%). However, low yields were obtained when diphenylamine and carbazole were used as the coupling partners with phenyl chloride, and the attempts to isolate the corresponding products failed.

Primary anilines are recognized as one of the poor coupling partners for the Buchwald–Hartwig amination reactions.[5a,13] Therefore, a series of primary anilines were also tested to examine the catalytic abilities of **C1**. As shown in Table 4, the results demonstrated that the arylations of the poorly nucleophilic primary anilines, including the sterically hindered ones, are all well tolerated, readily and selectively affording the desired monoarylated amines (**9a**, **9w**, **9y**, **10a**, **10w**, **13a**, **15a**, **15w**, and **16a**) in moderate to excellent yields (71–99%). In most cases, neither diarylation nor β-H elimination products were observed. Good results were also achieved regardless of the electronic properties of the substituted primary anilines. Additionally, for the primary alkylamine cyclohexanamine, good yields of monoarylated products (**9t**, **10t**, and **11t**) were achieved (84–91% yields).

Conclusions

In summary, a versatile method employing Pd–PEPPSI–IPr$^{(PhOMe)_2}$ (**C1**) was developed for the aryl ester and chloride Buchwald–Hartwig aminations under mild conditions. This method allows for a modular amide/amine synthesis under similar reaction conditions. The results demonstrated that the modification of the IPr backbone by introducing bulky and strong electron-donating dianisole groups resulted in a notable enhancement of the catalytic capabilities of Pd–PEPPSI–NHC complexes. Thus, the respective aminations of aryl esters and chlorides can be readily achieved with excellent chemoselectivities, high functional group tolerance, and broad substrate scope under mild conditions. Taking the combined scope of aryl esters and chlorides, the developed protocol offered a set of robust and general conditions for the Buchwald–Hartwig reactions. Efforts to further extend the scope of this transformation to other electrophiles for the Buchwald–Hartwig reactions are currently underway in our laboratory.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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Notes and references

1 For selected articles and reviews, see: (*a*) A. S. Guram, R. A. Rennels and S. L. Buchwald, *Angew. Chem., Int. Ed. Engl.*, 1995, **34**, 1348; (*b*) J. Louie and J. F. Hartwig, *Tetrahedron Lett.*, 1995, **36**, 3609; (*c*) J. F. Hartwig, *Angew. Chem., Int. Ed.*, 1998, **37**, 2046; (*d*) J. P. Wolfe, S. Wagaw, J.-F. Marcoux and S. L. Buchwald, *Acc. Chem. Res.*, 1998, **31**, 805; (*e*) M. M. Heravi, Z. Kheilkordi, V. Zadsirjan, M. Heydari and M. Malmir, *J. Organomet. Chem.*, 2018, **861**, 17; (*f*) P. A. Forero-Cortés and A. M. Haydl, *Org. Process Res. Dev.*, 2019, **23**, 1478; (*g*) R. Dorel, C. P. Grugel and A. M. Haydl, *Angew. Chem., Int. Ed.*, 2019, **58**, 17118.

2 (*a*) J. M. Dennis, N. A. White, R. Y. Liu and S. L. Buchwald, *J. Am. Chem. Soc.*, 2018, **140**, 4721; (*b*) T. Taeufer and J. Pospech, *J. Org. Chem.*, 2020, **85**, 7097.

3 Y. Zhang, G. Lavigne and V. César, *J. Org. Chem.*, 2015, **80**, 7666.

4 (*a*) W. Zhang and T. Nagashima, *J. Fluorine Chem.*, 2006, **127**, 588; (*b*) T. Lim, S. Byun and B. M. Kim, *Asian J. Org. Chem.*, 2017, **6**, 1222.

5 (*a*) T. Ben Halima, J. K. Vandavasi, M. Shkoor and S. G. Newman, *ACS Catal.*, 2017, **7**, 2176; (*b*) S. Shi and M. Szostak, *Chem. Commun.*, 2017, **53**, 10584; (*c*) T. Zhou, G. Li, S. P. Nolan and M. Szostak, *Org. Lett.*, 2019, **21**, 3304; (*d*) A. H. Dardir, P. R. Melvin, R. M. Davis, N. Hazari and M. Mohadjer Beromi, *J. Org. Chem.*, 2018, **83**, 469; (*e*) A. A. De la Fuente-Olvera, O. R. Suárez-Castillo and D. Mendoza-Espinosa, *Eur. J. Inorg. Chem.*, 2019, **2019**, 4879.

6 For selected reviews, see: (*a*) V. T. Abaev and O. V. Serdyuk, *Russ. Chem. Rev.*, 2008, **77**, 177; (*b*) A. S. Guram, *Org. Process Res. Dev.*, 2016, **20**, 1754; (*c*) E. A. Onoabedje and U. C. Okoro, *Synth. Commun.*, 2019, **49**, 2117For selected examples, see: (*d*) R. B. Bedford, C. S. J. Cazin, S. J. Coles, T. Gelbrich, P. N. Horton, M. B. Hursthouse and M. E. Light, *Organometallics*, 2003, **22**, 987; (*e*) C. V. Reddy, J. V. Kingston and J. G. Verkade, *J. Org. Chem.*, 2008, **73**, 3047; (*f*) R. J. Lundgren, A. Sappong-Kumankumah and M. Stradiotto, *Chem. – Eur. J.*, 2010, **16**, 1983; (*g*) A. Chartoire, M. Lesieur, A. M. Z. Slawin, S. P. Nolan and C. S. J. Cazin, *Organometallics*, 2011, **30**, 4432; (*h*) B. J. Tardiff, R. McDonald, M. J. Ferguson and M. Stradiotto, *J. Org. Chem.*, 2012, **77**, 1056; (*i*) C. A. Wheaton, J.-P. J. Bow and M. Stradiotto, *Organometallics*, 2013, **32**, 6148; (*j*) S. M. Crawford, C. B. Lavery and M. Stradiotto, *Chem. – Eur. J.*, 2013, **19**, 16760; (*k*) J. Mao, J. Zhang, S. Zhang and P. J. Walsh, *Dalton Trans.*, 2018, **47**, 8690; (*l*) M. Kim, T. Shin, A. Lee and H. Kim, *Organometallics*, 2018, **37**, 3253; (*m*) P. Weber, T. Scherpf, I. Rodstein, D. Lichte, L. T. Scharf, L. J. Gooßen and V. H. Gessner, *Angew. Chem., Int. Ed.*, 2019, **58**, 3203; (*n*) W. I. Lai, M. P. Leung, P. Y. Choy and F. Y. Kwong, *Synthesis*, 2019, **51**, 2678.

7 (*a*) P. G. Alsabeh and M. Stradiotto, *Angew. Chem., Int. Ed.*, 2013, **52**, 7242; (*b*) S. M. Wong, P. Y. Choy, O. Y. Yuen, C. M. So and F. Y. Kwong, *Org. Synth.*, 2015, **92**, 195.

8 For selected examples, see: (*a*) H. Tatamidani, K. Yokota, F. Kakiuchi and N. Chatani, *J. Org. Chem.*, 2004, **69**, 5615; (*b*) Y. Zhang and T. Rovis, *J. Am. Chem. Soc.*, 2004, **126**, 15964; (*c*) H. Tatamidani, F. Kakiuchi and N. Chatani, *Org. Lett.*, 2004, **6**, 3597; (*d*) K. W. Quasdorf, X. Tian and N. K. Garg, *J. Am. Chem. Soc.*, 2008, **130**, 14422; (*e*) B.-T. Guan, Y. Wang, B.-J. Li, D.-G. Yu and Z.-J. Shi, *J. Am. Chem. Soc.*, 2008, **130**, 14468; (*f*) B.-J. Li, Y.-Z. Li, X.-Y. Lu, J. Liu, B.-T. Guan and Z.-J. Shi, *Angew. Chem., Int. Ed.*, 2008, **47**, 10124; (*g*) N. A. LaBerge and J. A. Love, *Eur. J. Org. Chem.*, 2015, 5546; (*h*) T. Ben Halima, W. Zhang, I. Yalaoui, X. Hong, Y.-F. Yang, K. N. Houk and S. G. Newman, *J. Am. Chem. Soc.*, 2017, **139**, 1311; (*i*) C. A. Malapit, D. R. Caldwell, N. Sassu, S. Milbin and A. R. Howell, *Org. Lett.*, 2017, **19**, 1966; (*j*) P. Lei, G. Meng, S. Shi, Y. Ling, J. An, R. Szostak and M. Szostak, *Chem. Sci.*, 2017, **8**, 6525; (*k*) S. Shi, P. Lei and M. Szostak, *Organometallics*, 2017, **36**, 3784; (*l*) G. Li, S. Shi, P. Lei and M. Szostak, *Adv. Synth. Catal.*, 2018, **360**, 1538.

9 R. Takise, K. Muto and J. Yamaguchi, *Chem. Soc. Rev.*, 2017, **46**, 5864.

10 For selected reviews, see: (*a*) M. G. Organ, G. A. Chass, D.-C. Fang, A. C. Hopkinson and C. Valente, *Synthesis*, 2008, 2776; (*b*) C. Valente, M. E. Belowich, N. Hadei and M. G. Organ, *Eur. J. Org. Chem.*, 2010, 4343; (*c*) C. Valente, M. Pompeo, M. Sayah and M. G. Organ, *Org. Process Res. Dev.*, 2014, **18**, 180; (*d*) F. Izquierdo, S. Manzini and S. P. Nolan, *Chem. Commun.*, 2014, **50**, 14926.

11 For selected examples, see: (*a*) M. G. Organ, S. Çalimsiz, M. Sayah, K. H. Hoi and A. J. Lough, *Angew. Chem., Int. Ed.*, 2009, **48**, 2383; (*b*) S. Meiries, A. Chartoire, A. M. Z. Slawin and S. P. Nolan, *Organometallics*, 2012, **31**, 3402; (*c*) A. Chartoire, X. Frogneux, A. Boreux, A. M. Z. Slawin and S. P. Nolan, *Organometallics*, 2012, **31**, 6947; (*d*) T. Mitsui, M. Sugihara, Y. Tokoro and S.-I. Fukuzawa, *Tetrahedron*, 2015, **71**, 1509; (*e*) R. Maity, A. Verma, M. van der Meer, S. Hohloch and B. Sarkar, *Eur. J. Inorg. Chem.*, 2016, **2016**, 111; (*f*) X.-X. He, Y. Li, B.-B. Ma, Z. Ke and F.-S. Liu, *Organometallics*, 2016, **35**, 2655; (*g*) X.-B. Lan, F.-M. Chen, B.-B. Ma, D.-S. Shen and F.-S. Liu, *Organometallics*, 2016, **35**, 3852; (*h*) X.-B. Lan, Y. Li, Y.-F. Li, D.-S. Shen, Z. Ke and F.-S. Liu, *J. Org. Chem.*, 2017, **82**, 2914; (*i*) D.-D. Lu, X.-X. He and F.-S. Liu, *J. Org. Chem.*, 2017, **82**, 10898; (*j*) F.-D. Huang, C. Xu, D.-D. Lu, D.-S. Shen, T. Li and F.-S. Liu, *J. Org. Chem.*, 2018, **83**, 9144; (*k*) J.-S. Ouyang, Y.-F. Li, F.-D. Huang, D.-D. Lu and F.-S. Liu, *ChemCatChem*, 2018, **10**, 371; (*l*) F.-Y. Zhang, X.-B. Lan, C. Xu, H.-G. Yao, T. Li and F.-S. Liu, *Org. Chem. Front.*, 2019, **6**, 3292.

12 For selected examples, see: (*a*) T. Tu, W. Fang and J. Jiang, *Chem. Commun.*, 2011, **47**, 12358; (*b*) K. H. Hoi, J. A. Coggan and M. G. Organ, *Chem. – Eur. J.*, 2013, **19**, 843; (*c*) W. Fang, J. Jiang, Y. Xu, J. Zhou and T. Tu, *Tetrahedron*, 2013, **69**, 673; (*d*) M. Pompeo, J. L. Farmer, R. D. J. Froese and M. G. Organ, *Angew. Chem., Int. Ed.*, 2014, **53**, 3223; (*e*) Y. Zhang, V. César, G. Storch, N. Lugan and G. Lavigne, *Angew. Chem., Int. Ed.*, 2014, **53**, 6482; (*f*) S. Sharif, R. P. Rucker, N. Chandrasoma, D. Mitchell, M. J. Rodriguez, R. D. J. Froese and M. G. Organ, *Angew. Chem., Int. Ed.*, 2015, **54**, 9507; (*g*) Y. Zhang, V. César and G. Lavigne, *Eur. J. Org. Chem.*, 2015, 2042; (*h*) A. Winkler, K. Brandhorst, M. Freytag, P. G. Jones and M. Tamm, *Organometallics*, 2016, **35**, 1160; (*i*) D.-H. Li, X.-X. He, C. Xu, F.-D. Huang, N. Liu, D.-S. Shen and F.-S. Liu, *Organometallics*, 2019, **38**, 2539; (*j*) T. Mizusaki, K. Matsumoto, K. Takeuchi, N. Fukaya, Y. Takagi and J.-C. Choi, *Organometallics*, 2019, **38**, 1872.

13 (*a*) G. W. Cline and S. B. Hanna, *J. Am. Chem. Soc.*, 1987, **109**, 3087; (*b*) G. Wang, Q.-Y. Yu, J. Wang, S. Wang, S.-Y. Chen and X.-Q. Yu, *RSC Adv.*, 2013, **3**, 21306.